Microsoft Elevate

CAPSTONE PROJECT

IPL DATA ANALYSIS DASHBOARD

PRESENTED BY

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PROBLEM STATEMENT:

The Indian Premier League (IPL) generates a large amount of match and player performance data every season. However, this data is often scattered and difficult to analyze quickly . Cricket analysts, fans, and teams require a simple way to understand match statistics, player performance, team performance, and historical trends. Without proper visualization tools, it becomes difficult to derive meaningful insights from raw data . Therefore, there is a need for an interactive system that can analyze IPL data and present insights in an easy-to-understand visual format.

PROPOSED SOLUTION:

- The proposed system aims to develop an **interactive IPL Analysis Dashboard** to visualize and analyze IPL match data.
- The solution includes the following components:
- **Data Collection:**
  IPL datasets containing match details, team performance, player statistics, and venue information are collected from reliable online sources.
- **Data Cleaning & Preprocessing:**
  The dataset is cleaned by removing duplicate records, handling missing values, and organizing the data into a structured format suitable for analysis.
- **Data Visualization:**
  Power BI is used to create interactive dashboards showing key statistics such as total matches, top players, team performance, and match outcomes.
- **Interactive Analysis:**
  Users can filter the dashboard by team, season, or venue to analyze different performance trends.
- **Deployment:**
  The dashboard can be shared through Power BI reports or exported as a presentation for analysis and reporting purposes.

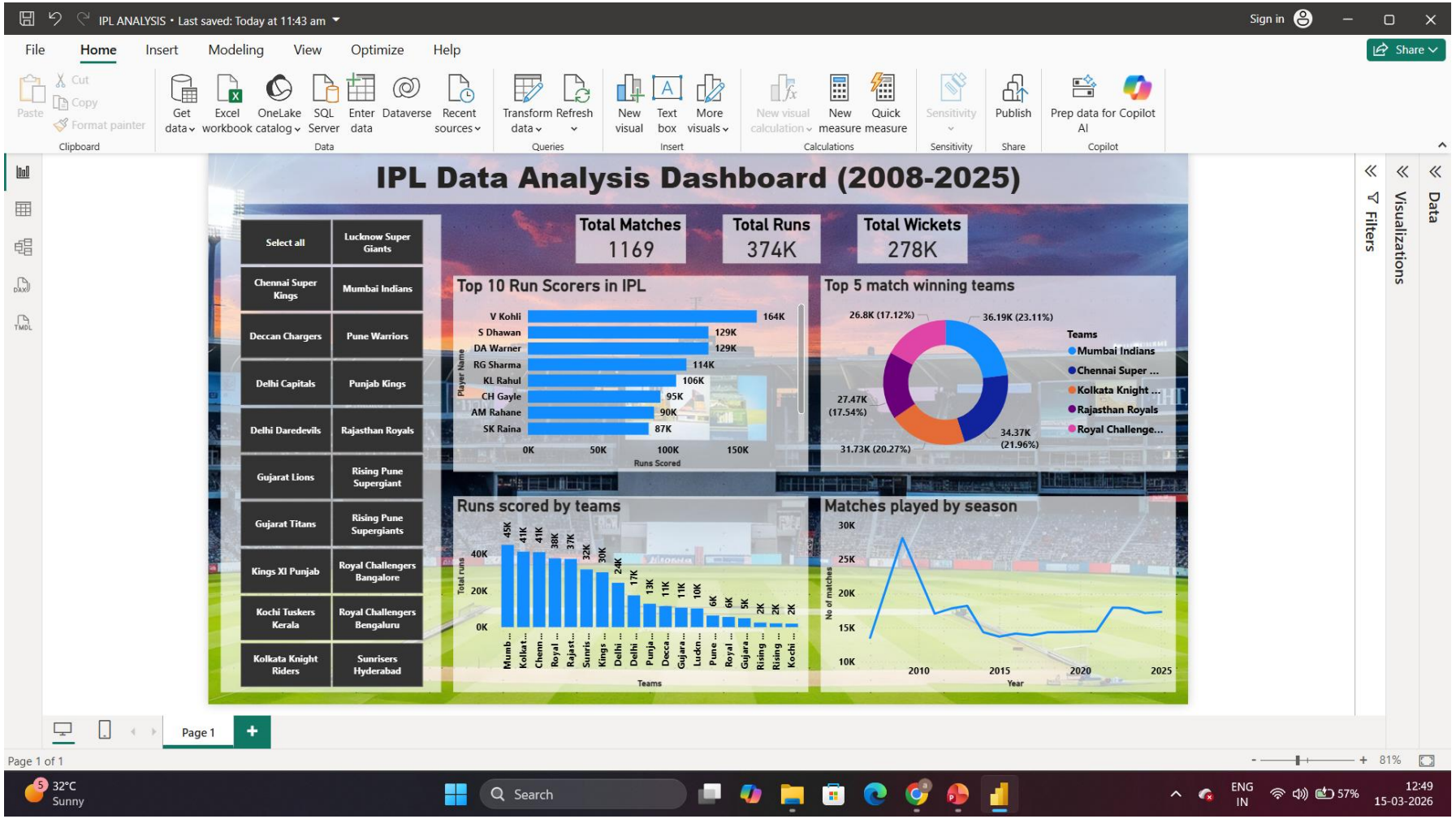
SYSTEM APPROACH:

- The system approach focuses on analysing IPL data using modern data visualization tools.
- **System Requirements**
- Hardware Requirements
  - Computer/Laptop
  - Minimum 4GB RAM
- Software Requirements
  - Microsoft Power BI Desktop
  - Microsoft Excel (for dataset preparation)
- **Libraries / Tools Used**
- • Microsoft Power BI
  - Data modelling tools in Power BI
  - DAX (Data Analysis Expressions)
  - Excel for data preprocessing
- These tools help in transforming raw IPL data into meaningful visual insights.

ALGORITHM & DEPLOYMENT:

- **Algorithm Selection**
- The system mainly uses **data analytics and visualization techniques** rather than machine learning algorithms.
- Power BI uses the following analytical processes:
- **Data Input**
- • IPL match dataset
  • Player statistics
  • Team performance data
  • Venue and season information
- **Data Processing**
- • Data cleaning and transformation using Power Query
  • Creating relationships between tables
  • Creating calculated measures using **DAX functions**
- **Deployment**
- The final dashboard is deployed in **Power BI Desktop**, where users can interact with visual reports and analyze IPL statistics dynamically.

RESULT:



The developed IPL Dashboard provides clear insights into IPL match statistics and team performance.

Key outputs include:

- Total matches played in IPL seasons
- Team-wise match wins analysis
- Top run scorers and top wicket takers
- Venue-wise match statistics
- Toss decision impact on match results

The dashboard displays the data through **interactive charts, bar graphs, pie charts, and KPI cards**, making the analysis easy to understand.

CONCLUSION:

The IPL Analysis Dashboard developed in this project demonstrates how data visualization tools like Microsoft Power BI can be used to transform large volumes of sports data into meaningful and interactive insights. By analyzing IPL datasets, the system provides a clear understanding of various match statistics such as team performance, player achievements, venue-based results, and season-wise trends. The dashboard enables users to easily explore and interpret complex data through visual elements like charts, graphs, and KPI indicators. This helps in identifying patterns such as the most successful teams, top-performing players, and the influence of factors like toss decisions or match venues on the outcome of games. Through this project, it is evident that data analytics plays an important role in sports management and performance analysis. The interactive nature of the dashboard allows users to filter and analyze data based on different parameters, making the analysis more efficient and user-friendly.

FUTURE SCOPE:

The system can be further improved with additional features such as:

- Integration of real-time IPL match data
- Use of machine learning models to predict match winners
- Player performance prediction
- Mobile dashboard for easy access
- Advanced analytics for team strategy planning

These improvements can make the system more powerful and useful for sports analysts and fans.

REFERENCES:

- IPL Dataset – Kaggle
- Microsoft Power BI Documentation
- Sports analytics research articles
- Microsoft Learn – Power BI Tutorials
- https://github.com/abhishek00006/IPL-PowerBI-Analysis

Thank You